PRE-ALGEBRA (8TH) · UNIT 5: PATTERNS AND BIVARIATE DATA

Screen Time & Sleep Study

Use scatter plots, lines of fit, and two-way tables to study screen time and sleep.

Honors

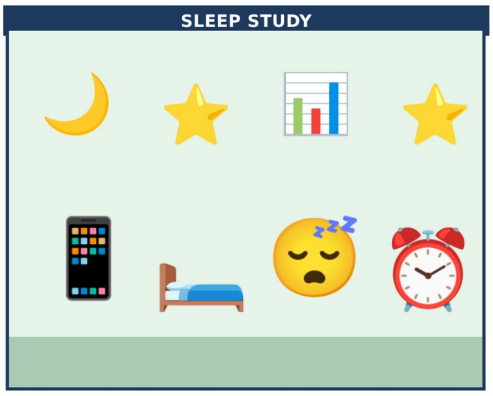
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

The school health committee wonders whether screen time is connected to how much students sleep. They collected data from students and want a clear, honest report.

You will analyze the data with scatter plots, a line of best fit, and a two-way table, and present what the data can and cannot tell us.



Model

sleep ≈ −0.45 × screen + 9.4

(hours per night)

Survey

50 students

phone in bedroom? sleeps 8+ h?

Learning Goals

- I can find and evaluate a linear model, including the point of averages.
- I can use a model to answer questions and judge its limits.
- I can analyze categorical data and design a study.

Project Timeline

When	What you will do
Day 1	Building the Model
Day 2	Using the Model
Day 3	Categorical Data
Final	Research Poster + reflection

Part 1: Building the Model

Day 1 · Use the Study Data card.

Study Data

Screen time (h) → Sleep (h)
1 → 9.0 · 2 → 8.6 · 2.5 → 8.4
3 → 8.0 · 4 → 7.6 · 4.5 → 7.5
5 → 7.0 · 6 → 6.6 · 6.5 → 6.4 · 8 → 5.8

Two-way table (50 students)
Phone in bedroom: 6 sleep 8+ h, 24 sleep < 8 h
No phone in bedroom: 14 sleep 8+ h, 6 sleep < 8 h

1. Write the line through (2, 8.6) and (6, 6.6). Compare it with y = −0.45x + 9.4.

2. Find the mean screen time and the mean sleep. Does the model pass close to (mean x, mean y)?

3. Find the residuals for x = 2 and x = 8 using the model.

Part 2: Using the Model

Day 2 · Use y = −0.45x + 9.4.

4. Doctors recommend at least 8 hours of sleep. According to the model, what is the most screen time that predicts 8 hours?

5. About how much less sleep does the model predict for a student with 6 hours of screen time than for one with 2 hours?

6. Describe an experiment that could test whether screen time causes less sleep.

Part 3: Categorical Data

Day 3 · Use the survey of 50 students.

7. Find the difference in the percent who sleep 8+ hours between the two groups.

8. If the whole school of 600 students had the same pattern, how many would have a phone in the bedroom and sleep 8+ hours?

9. Write your own survey question pair for a two-way table about sleep, and explain what result would show an association.

Final Product: Research Poster

Your final product must include:

- Model built from two points and checked against the mean point
- Residual analysis
- Model-based recommendation
- Experiment design
- Two-way analysis and a new survey proposal

Reflection

R1. Why does a good line of fit pass near the point of averages?

R2. How is a residual plot useful?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Scatter plot, model, predictions, and two-way analysis are accurate and correctly interpreted. Goes beyond the minimum with a creative, realistic choice.	Scatter plot, model, predictions, and two-way analysis are accurate and correctly interpreted.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Poster states findings clearly and explains limits (association vs. causation, extrapolation). Polished, organized, and visually engaging.	Poster states findings clearly and explains limits (association vs. causation, extrapolation). Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.